

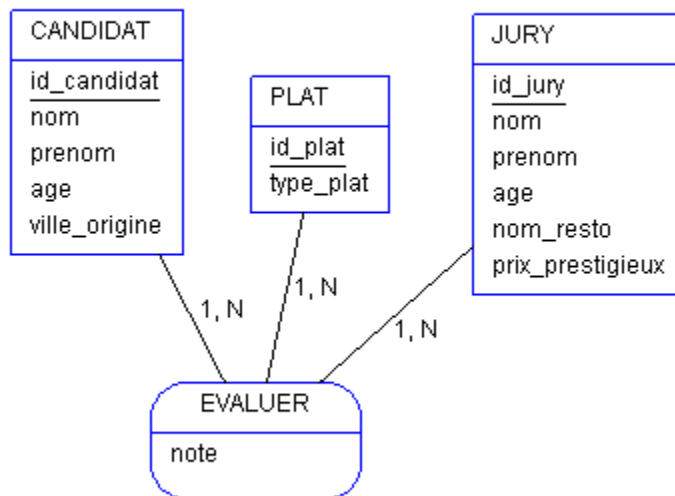
## **Bloc 2 TD Modélisation Avancée SQL**

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## Contexte

Ce TP va nous former à la modélisation et l'implémentation de bases de données, en appliquant cinq concepts complexes via des TD pratiques.

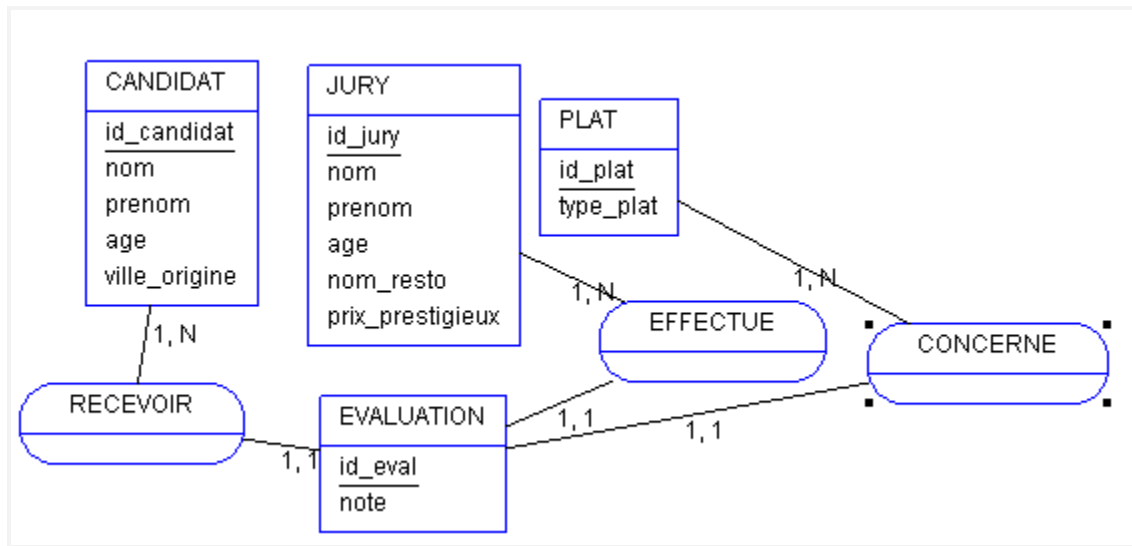
## Exercice 1



## Exercice 2

```
CANDIDAT (id_candidat_CANDIDAT, nom_CANDIDAT, prenom_CANDIDAT, age_CANDIDAT, ville_origine_CANDIDAT)
JURY (id_jury_JURY, nom_JURY, prenom_JURY, age_JURY, nom_resto_JURY, prix_prestigieux_JURY)
PLAT (id_plat_PLAT, type_plat_PLAT)
EVALUER (id_candidat_CANDIDAT, id_jury_JURY, id_plat_PLAT, note_EVALUER)
```

## Exercice 3



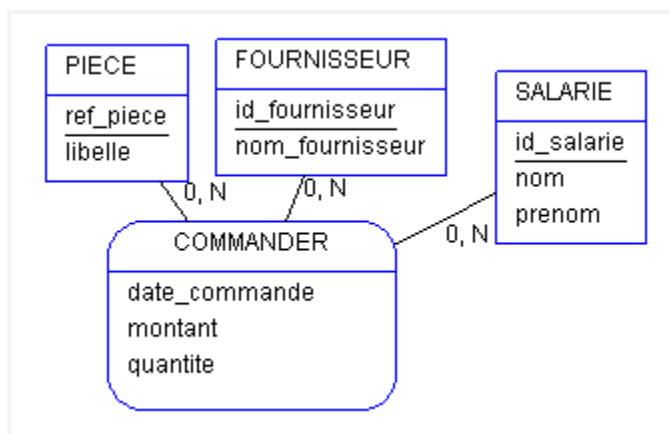
## Exercice 4

```

CANDIDAT (id_candidat_CANDIDAT, nom_CANDIDAT, prenom_CANDIDAT, age_CANDIDAT, ville_origine_CANDIDAT)
JURY (id_jury_JURY, nom_JURY, prenom_JURY, age_JURY, nom_resto_JURY, prix_prestigieux_JURY)
PLAT (id_plat_PLAT, type_plat_PLAT)
EVALUATION (id_eval_EVALUATION, note_EVALUER, #id_candidat_CANDIDAT, #id_jury_JURY, #id_plat_PLAT)

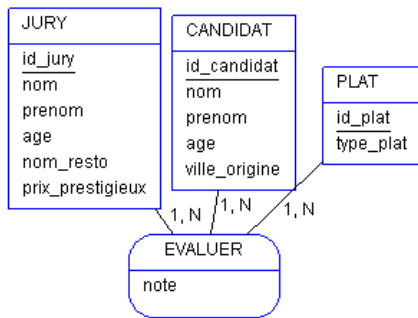
```

## Exercice 5



# TD 1

## MCD



## Requête SQL (Création des tables)

```
CREATE DATABASE IF NOT EXISTS ChefauTop;
USE ChefauTop;
```

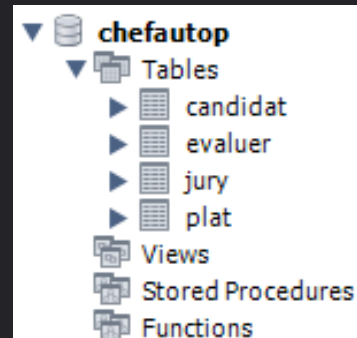
```
DROP TABLE IF EXISTS JURY ;
CREATE TABLE JURY (id_jury_JURY BIGINT(10) AUTO_INCREMENT NOT NULL,
nom_jury CHAR(50),
prenom_jury CHAR(50),
age_jury INT(3),
nom_resto_JURY CHAR(100),
prix_prestigieux_JURY CHAR(50),
PRIMARY KEY (id_jury_JURY)) ENGINE=InnoDB;

DROP TABLE IF EXISTS CANDIDAT ;
CREATE TABLE CANDIDAT (id_candidat_CANDIDAT BIGINT(10) AUTO_INCREMENT NOT NULL,
nom_cand CHAR(50),
prenom_cand CHAR(50),
age_cand INT(3),
ville_origine_CANDIDAT CHAR(100),
PRIMARY KEY (id_candidat_CANDIDAT)) ENGINE=InnoDB;

DROP TABLE IF EXISTS PLAT ;
CREATE TABLE PLAT (id_plat_PLAT BIGINT(10) AUTO_INCREMENT NOT NULL,
type_plat_PLAT CHAR(50),
PRIMARY KEY (id_plat_PLAT)) ENGINE=InnoDB;

DROP TABLE IF EXISTS EVALUER ;
CREATE TABLE EVALUER (id_jury_JURY BIGINT(10) NOT NULL,
id_candidat_CANDIDAT BIGINT(10) NOT NULL,
id_plat_PLAT BIGINT(10) NOT NULL,
note_EVALUER INT(10),
PRIMARY KEY (id_jury_JURY,
id_candidat_CANDIDAT,
id_plat_PLAT)) ENGINE=InnoDB;

ALTER TABLE EVALUER ADD CONSTRAINT FK_EVALUER_id_jury_JURY FOREIGN KEY (id_jury_JURY) REFERENCES JURY (id_jury_JURY);
ALTER TABLE EVALUER ADD CONSTRAINT FK_EVALUER_id_candidat_CANDIDAT FOREIGN KEY (id_candidat_CANDIDAT) REFERENCES CANDIDAT (id_candidat_CANDIDAT);
ALTER TABLE EVALUER ADD CONSTRAINT FK_EVALUER_id_plat_PLAT FOREIGN KEY (id_plat_PLAT) REFERENCES PLAT (id_plat_PLAT);
```



## Requête SQL (Insertion des éléments)

```
INSERT INTO PLAT (id_plat_PLAT, type_plat_PLAT) VALUES
(1, 'Entrée'),
(2, 'Plat principal'),
(3, 'dessert');

INSERT INTO CANDIDAT (id_candidat_CANDIDAT, nom_cand, prenom_cand) VALUES
(1, 'touille', 'rata'),
(2, 'L''éponge', 'Bob'),
(3, 'Man', 'Spider');

INSERT INTO JURY (id_jury_JURY, nom_jury, prenom_jury) VALUES
(1, 'Gravouil', 'Benjamin'),
(2, 'Dindar', 'Djamil'),
(3, 'Ducasse', 'Alain'),
(4, 'Bo', 'Harry'),
(5, 'Tout', 'Bouffe');

INSERT INTO EVALUER (id_candidat_CANDIDAT, id_plat_PLAT, id_jury_JURY, note_EVALUER) VALUES
-- Candidat 1
(1, 1, 1, 12), (1, 1, 2, 15), (1, 1, 3, 11), (1, 1, 4, 13), (1, 1, 5, 18),
(1, 2, 1, 11), (1, 2, 2, 12), (1, 2, 3, 4), (1, 2, 4, 11), (1, 2, 5, 17),
(1, 3, 1, 15), (1, 3, 2, 15), (1, 3, 3, 15), (1, 3, 4, 13), (1, 3, 5, 14),
-- Candidat 2
(2, 1, 1, 18), (2, 1, 2, 14), (2, 1, 3, 12), (2, 1, 4, 11), (2, 1, 5, 18),
(2, 2, 1, 11), (2, 2, 2, 13), (2, 2, 3, 4), (2, 2, 4, 11), (2, 2, 5, 16),
(2, 3, 1, 12), (2, 3, 2, 12), (2, 3, 3, 8), (2, 3, 4, 13), (2, 3, 5, 20),
-- Candidat 3
(3, 1, 1, 18), (3, 1, 2, 12), (3, 1, 3, 5), (3, 1, 4, 7), (3, 1, 5, 18),
(3, 2, 1, 9), (3, 2, 2, 14), (3, 2, 3, 4), (3, 2, 4, 12), (3, 2, 5, 16),
(3, 3, 1, 11), (3, 3, 2, 12), (3, 3, 3, 12), (3, 3, 4, 14), (3, 3, 5, 20);
```

## Requête SQL (Affichage d'information)

Affichage du nombre total de notes :

```
1  SELECT COUNT(note_EVALUER)
2  FROM EVALUER;
```

|   | COUNT(note_EVALUER) |
|---|---------------------|
| ▶ | 45                  |

Affichage des moyennes des candidats :

```
SELECT nom_cand, prenom_cand, AVG(note_EVALUER)
FROM CANDIDAT, EVALUER
WHERE CANDIDAT.id_candidat_CANDIDAT = EVALUER.id_candidat_CANDIDAT
GROUP BY nom_cand, prenom_cand;
```

|   | nom_cand | prenom_cand | AVG(note_EVALUER) |
|---|----------|-------------|-------------------|
| ▶ | touille  | rata        | 13.0667           |
|   | L'éponge | Bob         | 12.8667           |
|   | Man      | Spider      | 12.2667           |

Affichage des moyennes trié dans l'ordre croissant :

```
SELECT nom_cand, prenom_cand, AVG(note_EVALUER) AS moyenne
FROM CANDIDAT, EVALUER
WHERE CANDIDAT.id_candidat_CANDIDAT = EVALUER.id_candidat_CANDIDAT
GROUP BY nom_cand, prenom_cand ORDER BY moyenne ASC;
```

|   | nom_cand | prenom_cand | moyenne |
|---|----------|-------------|---------|
| ▶ | Man      | Spider      | 12.2667 |
|   | L'éponge | Bob         | 12.8667 |
|   | touille  | rata        | 13.0667 |

Affichage du vainqueur (la meilleur note) :

```
SELECT nom_cand, prenom_cand, AVG(note_EVALUER) AS moyenne
FROM CANDIDAT, EVALUER
WHERE CANDIDAT.id_candidat_CANDIDAT = EVALUER.id_candidat_CANDIDAT
GROUP BY nom_cand, prenom_cand
ORDER BY moyenne DESC
LIMIT 1;
```

|   | nom_cand | prenom_cand | moyenne |
|---|----------|-------------|---------|
| ▶ | touille  | rata        | 13.0667 |

Affiche des moyennes données par les jury

```
SELECT nom_jury, prenom_jury, AVG(note_EVALUER)
FROM JURY, EVALUER
WHERE JURY.id_jury_JURY = EVALUER.id_jury_JURY
GROUP BY nom_jury, prenom_jury
```

|   | nom_jury | prenom_jury | AVG(note_EVALUER) |
|---|----------|-------------|-------------------|
| ▶ | Gravouil | Benjamin    | 13.0000           |
|   | Dindar   | Djamil      | 13.2222           |
|   | Ducasse  | Alain       | 8.3333            |
|   | Bo       | Harry       | 11.6667           |
|   | Tout     | Bouffe      | 17.4444           |

Affichage des moyennes trié par les plus sévère par les jurys :

```
SELECT nom_jury, prenom_jury, AVG(note_EVALUER) AS moyenne
FROM JURY, EVALUER
WHERE JURY.id_jury_JURY = EVALUER.id_jury_JURY
GROUP BY nom_jury, prenom_jury ORDER BY moyenne ASC;
```

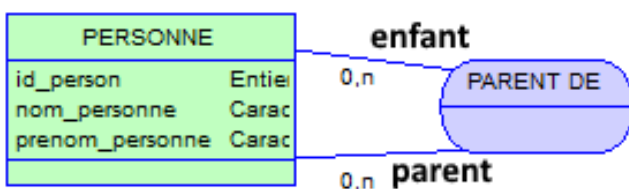
|   | nom_jury | prenom_jury | moyenne |
|---|----------|-------------|---------|
| ► | Ducasse  | Alain       | 8.3333  |
|   | Bo       | Harry       | 11.6667 |
|   | Gravouil | Benjamin    | 13.0000 |
|   | Dindar   | Djamil      | 13.2222 |
|   | Tout     | Bouffe      | 17.4444 |

Affichage des moyennes trié par les plus clémentes par les jurys :

```
SELECT nom_jury, prenom_jury, AVG(note_EVALUER) AS moyenne
FROM JURY, EVALUER
WHERE JURY.id_jury_JURY = EVALUER.id_jury_JURY
GROUP BY nom_jury, prenom_jury ORDER BY moyenne DESC;
```

|  | nom_jury | prenom_jury | moyenne |
|--|----------|-------------|---------|
|  | Tout     | Bouffe      | 17.4444 |
|  | Dindar   | Djamil      | 13.2222 |
|  | Gravouil | Benjamin    | 13.0000 |
|  | Bo       | Harry       | 11.6667 |
|  | Ducasse  | Alain       | 8.3333  |

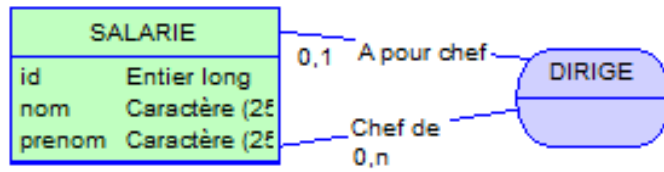
## Exercice 6



PERSONNE(id\_person, nom\_personne, prenom\_personne)

PARENT\_DE(#id\_parent, #id\_enfant, quantite) -> id\_parent\_PERSONNE

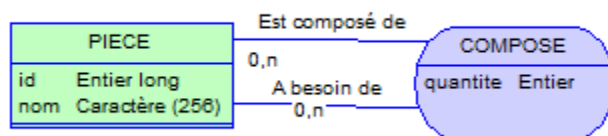
## Exercice 7



SALARIE(id, nom, prenom, #id\_chef)

#id\_chef -> id de la table SALARIE

## Exercice 8



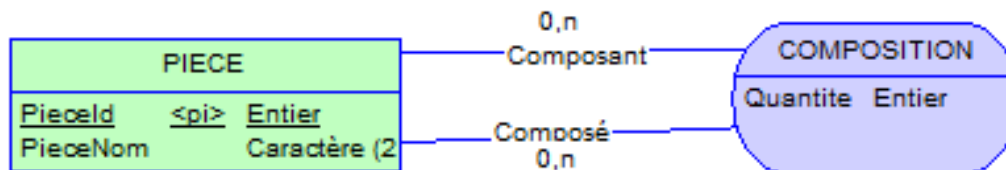
PIÈCE(id, nom)

COMPOSE(#id\_besoin, #id\_composer, quantite) -> id\_PIÈCE



# TD 2

## MCD



## Préparation du MPD :

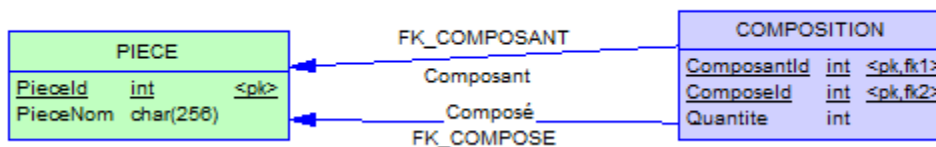
Propriétés de la table - COMPOSITION (COMPOSITION)

Correspondances | Notes | Règles | Dépendances | Dépendances étendues | Version

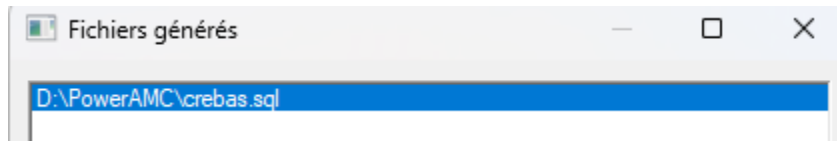
Général | Colonnes | Index | Clés | Contrôle | Script | Options | Aperçu

|   | Nom         | Code        | Type de do | P | E | O |
|---|-------------|-------------|------------|---|---|---|
| 1 | ComposantId | COMPOSANTID | int        | ✓ | ✓ | ✓ |
| → | ComposeId   | COMPOSEID   | int        | ✓ | ✓ | ✓ |
| 3 | Quantite    | QUANTITE    | int        |   |   |   |

OK Annuler Appliquer Aide



## Génération BDD :



```
/*=====*/
/* Nom de la base : SLAM1_TD2_AVION */
/* Nom de SGBD : MySQL 4.0 */
/* Date de création : 23/03/2026 13:27:55 */
/*=====*/

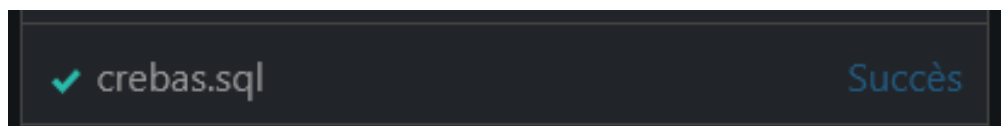
use SLAM1_TD2_AVION;

drop database if exists SLAM1_TD2_AVION;

/*=====*/
/* Base de donnée : SLAM1_TD2_AVION */
/*=====*/
create database SLAM1_TD2_AVION;

use SLAM1_TD2_AVION;
```

Remplacement de “type = InnoDB;” par “ENGINE = InnoDB” car type est obsolète (avant MySQL 5.5)



|                          |             |   |   |           |   |           |   |            |   |         |    |       |   |           |   |        |                    |          |   |
|--------------------------|-------------|---|---|-----------|---|-----------|---|------------|---|---------|----|-------|---|-----------|---|--------|--------------------|----------|---|
| <input type="checkbox"/> | composition | ☆ | ■ | Parcourir | 📄 | Structure | 🔍 | Rechercher | ➕ | Insérer | 🗑️ | Vider | ➖ | Supprimer | 0 | InnoDB | utf8mb4_general_ci | 48,0 kio | - |
| <input type="checkbox"/> | piece       | ☆ | ■ | Parcourir | 📄 | Structure | 🔍 | Rechercher | ➕ | Insérer | 🗑️ | Vider | ➖ | Supprimer | 0 | InnoDB | utf8mb4_general_ci | 16,0 kio | - |

Execution SQL pour correspondre au TD :

```
DROP TABLE IF EXISTS COMPOSITION;
DROP TABLE IF EXISTS PIECE;

CREATE TABLE PIECE
(
    PIECEID          VARCHAR(50)          NOT NULL,
    PIECENOM         VARCHAR(200),
    PRIMARY KEY (PIECEID)
)
ENGINE = InnoDB;

CREATE TABLE COMPOSITION
(
    COMPOSANTID      VARCHAR(50)          NOT NULL,
    COMPOSEID        VARCHAR(50)          NOT NULL,
    QUANTITE         INT,
    PRIMARY KEY (COMPOSANTID, COMPOSEID)
)
ENGINE = InnoDB;

ALTER TABLE COMPOSITION ADD CONSTRAINT FK_COMPOSANT FOREIGN KEY (COMPOSEID)
REFERENCES PIECE (PIECEID) ON DELETE RESTRICT ON UPDATE RESTRICT;

ALTER TABLE COMPOSITION ADD CONSTRAINT FK_COMPOSE FOREIGN KEY (COMPOSANTID)
REFERENCES PIECE (PIECEID) ON DELETE RESTRICT ON UPDATE RESTRICT;

INSERT INTO PIECE (PIECEID, PIECENOM) VALUES
('aile', 'Aile d''avion'),
('aileron', 'Aileron d''aile'),
('longeron', 'Longeron d''aile'),
('charnière', 'Charnière à tout faire'),
('rivet', 'rivets à la pelle'),
('train', 'Train de machin');

INSERT INTO COMPOSITION (COMPOSANTID, COMPOSEID, QUANTITE) VALUES
('aile', 'aileron', 1),
('aile', 'longeron', 5),
('aile', 'rivet', 100),
('aile', 'train', 1),
('aileron', 'charnière', 2),
('aileron', 'rivet', 5),
('charnière', 'rivet', 4),
('longeron', 'rivet', 10),
('train', 'charnière', 3),
('train', 'rivet', 8);
```

## Requête SQL (Affichage d'information)

Affichage du nom et de la quantité de pièces à commander pour une aile (on ignore les sous pièces)

```
SELECT PIECE.PieceNom, COMPOSITION.Quantite FROM  
COMPOSITION, PIECE WHERE COMPOSITION.ComposeId =  
PIECE.PieceId AND COMPOSITION.ComposantId = 'aile';
```

| PieceNom          | Quantite |
|-------------------|----------|
| Aileron d'aile    | 1        |
| Longeron d'aile   | 5        |
| rivets à la pelle | 100      |
| Train de machin   | 1        |

Affichage du nombre de pièce à commander pour une aile (on ignore les sous pièces)

```
SELECT SUM(Quantite) AS NombreTotalPieces FROM COMPOSITION  
WHERE ComposantId = 'aile';
```

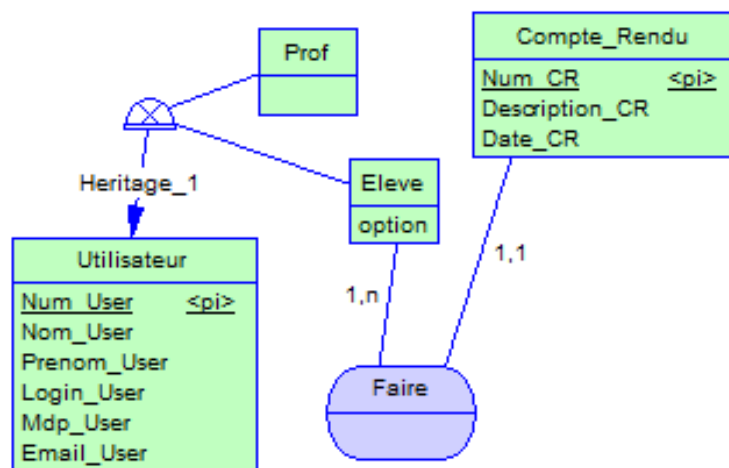
| NombreTotalPieces |
|-------------------|
| 107               |

Affichage du nom et de la quantité des pièces à commander pour une train de machin (on ignore les sous pièces) (on ignore l'id de la pièce)

```
SELECT PIECE.PieceNom, COMPOSITION.Quantite FROM  
COMPOSITION, PIECE WHERE COMPOSITION.ComposeId =  
PIECE.PieceId AND COMPOSITION.ComposantId = ( SELECT PieceId  
FROM PIECE WHERE PieceNom = 'Train de machin' );
```

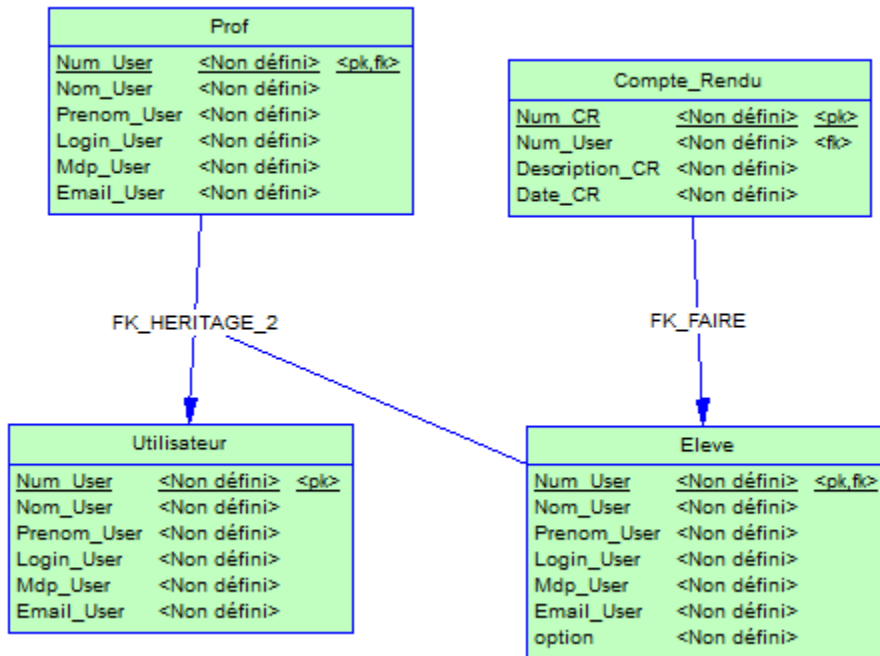
| PieceNom               | Quantite |
|------------------------|----------|
| Charnière à tout faire | 3        |
| rivets à la pelle      | 8        |

## Exercice 9

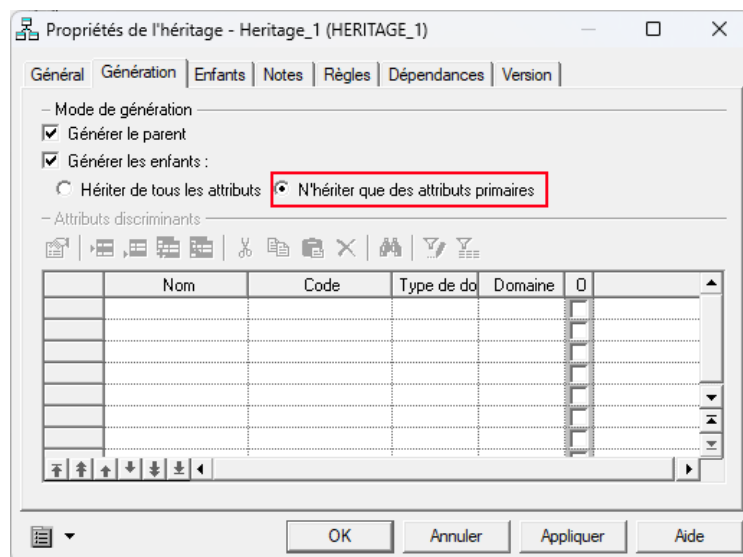


## Exercice 10

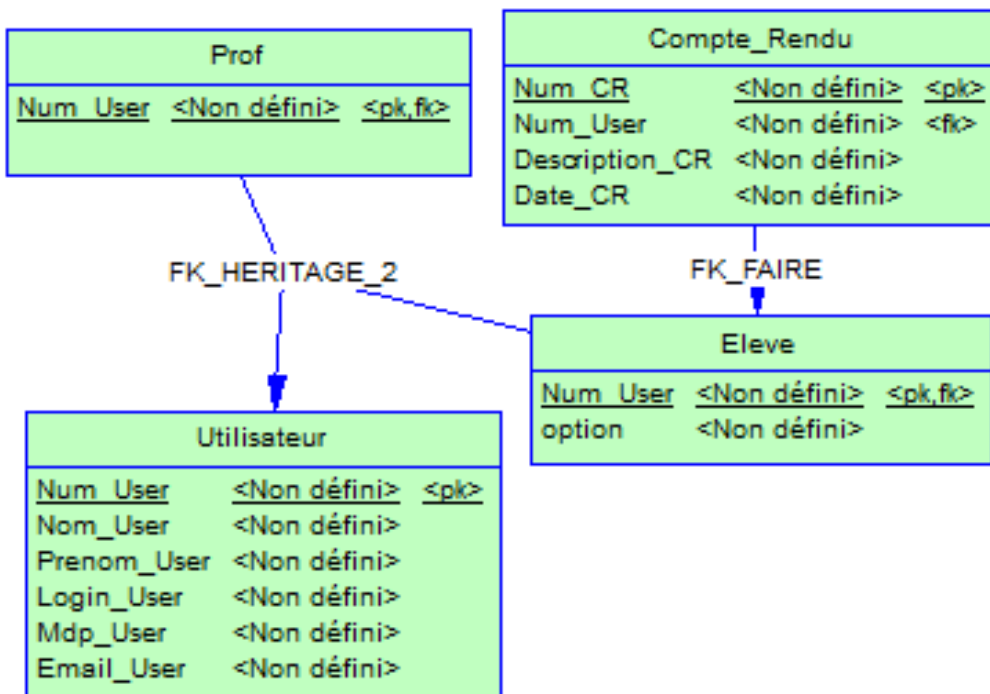
MPD généré par PowerAMC :



Mais avec cette modification :



On obtient ça :



# TD 3

## Insertion SQL dans la BDD :

```
drop table if exists COMPTE_RENDU;
drop table if exists ELEVE;
drop table if exists PROF;
drop table if exists UTILISATEUR;

create table COMPTE_RENDU
(
    NUM_CR                int                not null,
    NUM_USER              int                not null,
    DESCRIPTION_CR        char(100),
    DATE_CR               Date,
    primary key (NUM_CR)
)

ENGINE = InnoDB;

create index FAIRE_FK on COMPTE_RENDU
(
    NUM_USER
);

create table ELEVE
(
    NUM_USER              int                not null,
    `OPTION`              char(10),
    primary key (NUM_USER)
)
ENGINE = InnoDB;

create table PROF
(
    NUM_USER              int                not null,
    primary key (NUM_USER)
)
ENGINE = InnoDB;

create table UTILISATEUR
(
    NUM_USER              int                not null,
    NOM_USER              char(50),
    PRENOM_USER           char(50),
    LOGIN_USER            char(50),
    MDP_USER              char(100),
    EMAIL_USER            char(100),
    primary key (NUM_USER)
)
ENGINE = InnoDB;

alter table COMPTE_RENDU add constraint FK_FAIRE foreign key (NUM_USER)
references ELEVE (NUM_USER) on delete restrict on update restrict;

alter table ELEVE add constraint FK_HERITAGE_1 foreign key (NUM_USER)
references UTILISATEUR (NUM_USER) on delete restrict on update restrict;

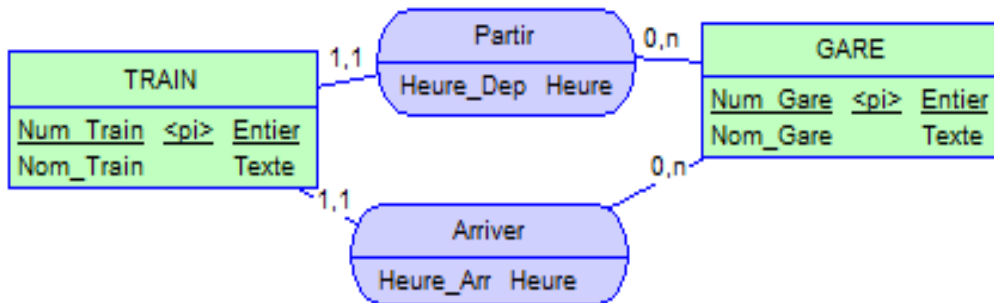
alter table PROF add constraint FK_HERITAGE_2 foreign key (NUM_USER)
references UTILISATEUR (NUM_USER) on delete restrict on update restrict;
```

| Table ↕                                      | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Lignes ① | Type   | Interclassement    | Taille   | Perte |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|--------------------|----------|-------|
| <input type="checkbox"/> <b>compte_rendu</b> | ☆       | 0        | InnoDB | utf8mb4_general_ci | 32,0 kio | -     |
| <input type="checkbox"/> <b>eleve</b>        | ☆       | 0        | InnoDB | utf8mb4_general_ci | 16,0 kio | -     |
| <input type="checkbox"/> <b>prof</b>         | ☆       | 0        | InnoDB | utf8mb4_general_ci | 16,0 kio | -     |
| <input type="checkbox"/> <b>utilisateur</b>  | ☆       | 0        | InnoDB | utf8mb4_general_ci | 16,0 kio | -     |
| 4 tables                                     | Somme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0        | InnoDB | utf8mb4_general_ci | 80,0 kio | 0 o   |

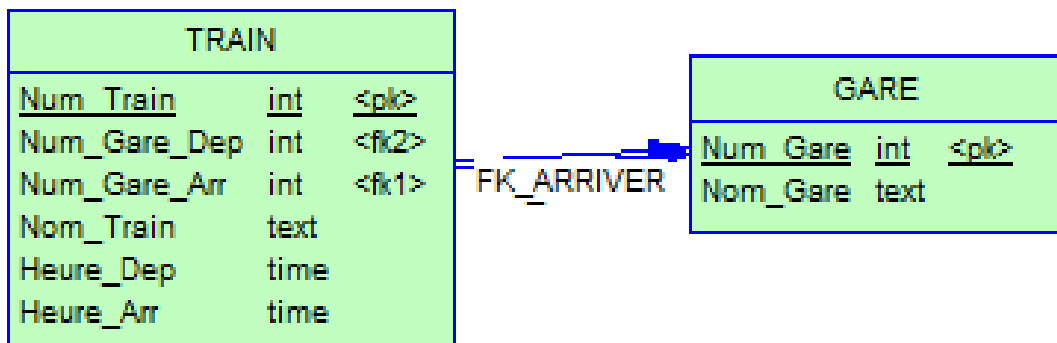


# Exercise 11

MCD :

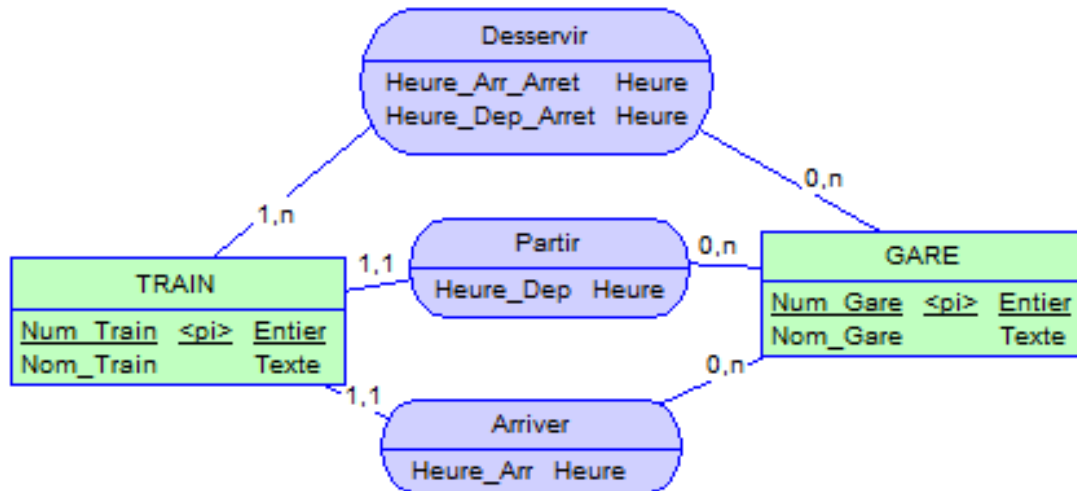


MPD (limitation PowerAMC) :

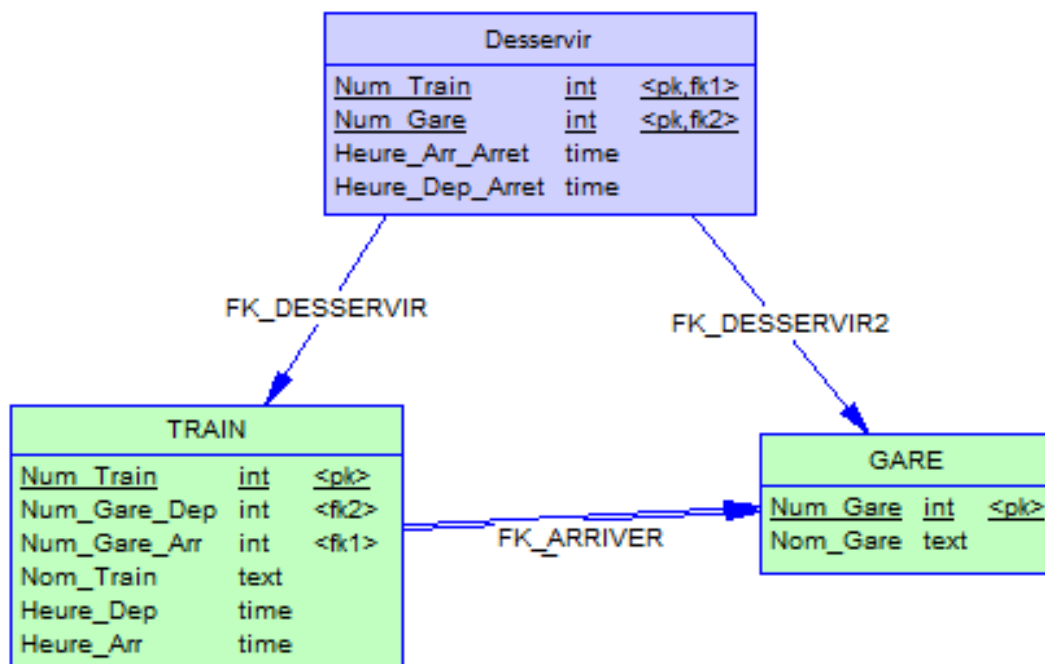


## Exercice 12

Nouveau MCD :

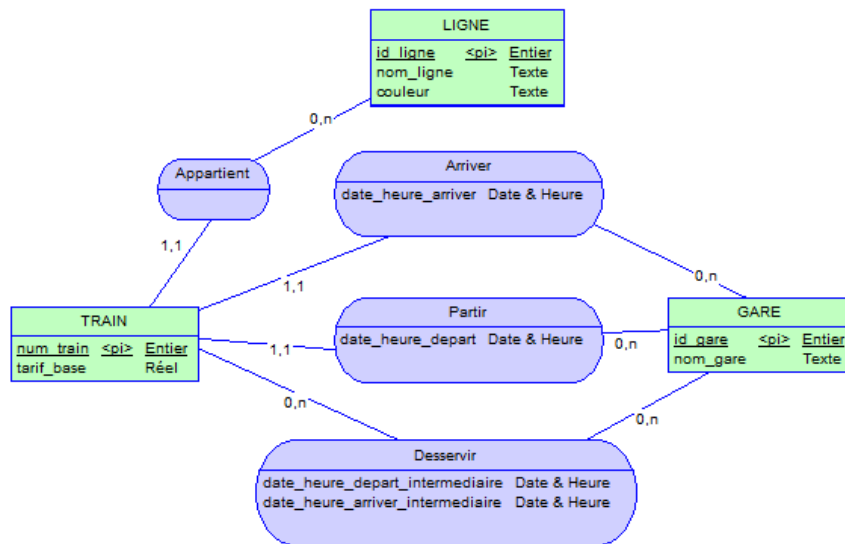


Nouveau MPD (limitation PowerAMC) :

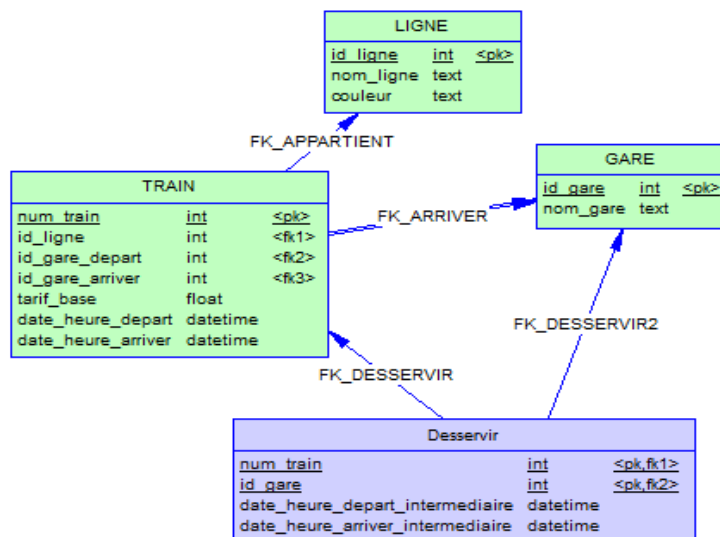


# TD 4

## MCD



## Préparation du MPD



## Génération BDD :

Remplacement de “type = InnoDB;” par “ENGINE = InnoDB” car type est obsolète (avant MySQL 5.5)

```
drop table if exists DESSERVIR;
drop table if exists GARE;
drop table if exists LIGNE;
drop table if exists TRAIN;
create table DESSERVIR
(
    NUM_TRAIN          int          not null,
    ID_GARE             int          not null,
    DATE_HEURE_DEPART_INTERMEDIAIRE datetime,
    DATE_HEURE_ARRIVER_INTERMEDIAIRE datetime,
    primary key (NUM_TRAIN, ID_GARE)
)
ENGINE = InnoDB;
create index DESSERVIR_FK on DESSERVIR
(
    NUM_TRAIN
);
create index DESSERVIR2_FK on DESSERVIR
(
    ID_GARE
);
create table GARE
(
    ID_GARE             int          not null,
    NOM_GARE            text,
    primary key (ID_GARE)
)
ENGINE = InnoDB;
create table LIGNE
(
    ID_LIGNE            int          not null,
    NOM_LIGNE           text,
    COULEUR             text,
    primary key (ID_LIGNE)
)
ENGINE = InnoDB;
create table TRAIN
(
    NUM_TRAIN           int          not null,
    ID_LIGNE            int          not null,
    ID_GARE_DEPART      int          not null,
    ID_GARE_ARRIVER     int          not null,
    TARIF_BASE          float,
    DATE_HEURE_DEPART   datetime,
    DATE_HEURE_ARRIVER  datetime,
    primary key (NUM_TRAIN)
)
ENGINE = InnoDB;
create index APPARTIENT_FK on TRAIN
(
    ID_LIGNE
);
```

```

create index PARTIR_FK on TRAIN
(
    ID_GARE_DEPART
);
create index ARRIVER_FK on TRAIN
(
    ID_GARE_ARRIVER
);
alter table DESSERVIR add constraint FK_DESSERVIR foreign key (NUM_TRAIN)
references TRAIN (NUM_TRAIN) on delete restrict on update restrict;
alter table DESSERVIR add constraint FK_DESSERVIR2 foreign key (ID_GARE)
references GARE (ID_GARE) on delete restrict on update restrict;
alter table TRAIN add constraint FK_APPARTIENT foreign key (ID_LIGNE)
references LIGNE (ID_LIGNE) on delete restrict on update restrict;
alter table TRAIN add constraint FK_ARRIVER foreign key (ID_GARE_ARRIVER)
references GARE (ID_GARE) on delete restrict on update restrict;
alter table TRAIN add constraint FK_PARTIR foreign key (ID_GARE_DEPART)
references GARE (ID_GARE) on delete restrict on update restrict;

```

Création de la BDD dans PhpMyAdmin :



Importation du fichier .sql dans la BDD :

| Table                              | Action                                                   | Lignes | Type   | Interclassement    | Taille    | Perte |
|------------------------------------|----------------------------------------------------------|--------|--------|--------------------|-----------|-------|
| <input type="checkbox"/> desservir | ☆ Parcourir Structure Rechercher Insérer Vider Supprimer | 0      | InnoDB | utf8mb4_general_ci | 48,0 kio  | -     |
| <input type="checkbox"/> gare      | ☆ Parcourir Structure Rechercher Insérer Vider Supprimer | 0      | InnoDB | utf8mb4_general_ci | 16,0 kio  | -     |
| <input type="checkbox"/> ligne     | ☆ Parcourir Structure Rechercher Insérer Vider Supprimer | 0      | InnoDB | utf8mb4_general_ci | 16,0 kio  | -     |
| <input type="checkbox"/> train     | ☆ Parcourir Structure Rechercher Insérer Vider Supprimer | 0      | InnoDB | utf8mb4_general_ci | 64,0 kio  | -     |
| 4 tables                           | Somme                                                    | 0      | InnoDB | utf8mb4_general_ci | 144,0 kio | 0 o   |

## Requête SQL (Insertion des éléments)

Nous devons insérer ces informations :

**étape 7** : Insérer dans votre base de données le train n°242:

- ligne n°2 - vert : Simplon-Orient-Express
- départ de Paris le 10/03/2020 à 8h30
- arrivée à Istanbul le 15/03/2020 à 16h45
- tarif : 4000€
- gares desservies :
  - Lausanne : le 10/03/2020 à 16h15
  - Milan : le 11/03/2020 à 9h45
  - Venise : le 12/03/2020 à 8h
  - Belgrade : le 13/03/2020 à 19h55
  - Sofia : le 14/03/2020 à 16h

**étape 8** : Insérer dans votre base de données le train n°422:

- ligne n°4 - violet : Venise-Simplon-Orient-Express
- départ de Calais le 10/03/2020 à 9h00
- arrivée à Venise le 12/03/2020 à 20h30
- tarif : 2360 €
- gares desservies :
  - Paris : le 10/03/2020 à 12h
  - Lausanne : le 11/03/2020 à 8h
  - Milan : le 11/03/2020 à 21h

La table **DESSERVIR** exige une heure d'arrivée et de départ, je vais considérer l'heure d'arrivée comme étant de 15 minutes avant l'heure de départ.

Insertion du référentiel :

```
INSERT INTO LIGNE (id_ligne, nom_ligne, couleur) VALUES (2, 'Simplon-Orient-Express', 'vert'), (4, 'Venise-Simplon-Orient-Express', 'violet');
```

```
INSERT INTO GARE (id_gare, nom_gare) VALUES (1, 'Paris'), (2, 'Istanbul'), (3, 'Lausanne'), (4, 'Milan'), (5, 'Venise'), (6, 'Belgrade'), (7, 'Sofia'), (8, 'Calais');
```

Ajout du train n°242 :

```
INSERT INTO TRAIN (num_train, tarif_base, id_ligne,  
id_gare_depart, id_gare_arriver, date_heure_depart,  
date_heure_arriver) VALUES (242, 4000.00, 2, 1, 2,  
'2020-03-10 08:30:00', '2020-03-15 16:45:00');
```

```
INSERT INTO Desservir (num_train, id_gare,  
date_heure_arriver_intermediaire,  
date_heure_depart_intermediaire) VALUES (242, 3, '2020-03-10  
16:00:00', '2020-03-10 16:15:00'), (242, 4, '2020-03-11  
09:30:00', '2020-03-11 09:45:00'), (242, 5, '2020-03-12  
07:45:00', '2020-03-12 08:00:00'), (242, 6, '2020-03-13  
19:40:00', '2020-03-13 19:55:00'), (242, 7, '2020-03-14  
15:45:00', '2020-03-14 16:00:00');
```

Ajout du train n°422 :

```
INSERT INTO TRAIN (num_train, tarif_base, id_ligne,  
id_gare_depart, id_gare_arriver, date_heure_depart,  
date_heure_arriver) VALUES (422, 2360.00, 4, 8, 5,  
'2020-03-10 09:00:00', '2020-03-12 20:30:00');
```

```
INSERT INTO Desservir (num_train, id_gare,  
date_heure_arriver_intermediaire,  
date_heure_depart_intermediaire) VALUES (422, 1, '2020-03-10  
11:45:00', '2020-03-10 12:00:00'), (422, 3, '2020-03-11  
07:45:00', '2020-03-11 08:00:00'), (422, 4, '2020-03-11  
20:45:00', '2020-03-11 21:00:00');
```

## Requête SQL (Vérification des éléments)

```
SELECT t.num_train, l.nom_ligne, g_dep.nom_gare AS depart,
t.date_heure_depart, g_arr.nom_gare AS arriver,
t.date_heure_arriver, t.tarif_base FROM TRAIN t, LIGNE l,
GARE g_dep, GARE g_arr WHERE t.id_ligne = l.id_ligne AND
t.id_gare_depart = g_dep.id_gare AND t.id_gare_arriver =
g_arr.id_gare;
```

| num_train | nom_ligne                     | depart | date_heure_depart   | arriver  | date_heure_arriver  | tarif_base |
|-----------|-------------------------------|--------|---------------------|----------|---------------------|------------|
| 242       | Simplon-Orient-Express        | Paris  | 2020-03-10 08:30:00 | Istanbul | 2020-03-15 16:45:00 | 4000       |
| 422       | Venise-Simplon-Orient-Express | Calais | 2020-03-10 09:00:00 | Venise   | 2020-03-12 20:30:00 | 2360       |

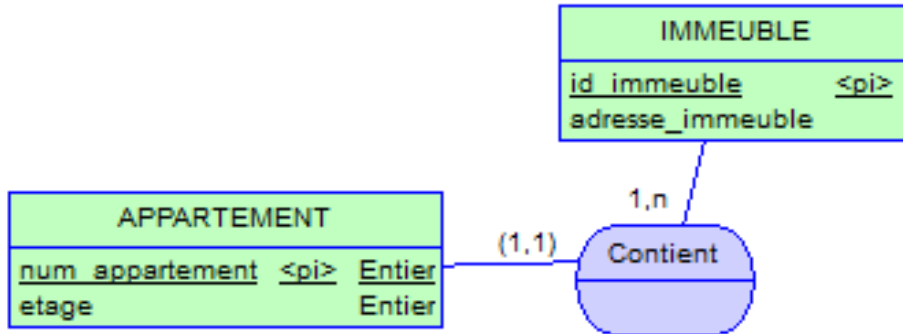
```
SELECT d.num_train, g.nom_gare AS gare_intermediaire,
d.date_heure_arriver_intermediaire,
d.date_heure_depart_intermediaire FROM Desservir d, GARE g
WHERE d.id_gare = g.id_gare ORDER BY d.num_train,
d.date_heure_arriver_intermediaire;
```

| num_train | gare_intermediaire | date_heure_arriver_intermediaire | date_heure_depart_intermediaire |
|-----------|--------------------|----------------------------------|---------------------------------|
| 242       | Lausanne           | 2020-03-10 16:00:00              | 2020-03-10 16:15:00             |
| 242       | Milan              | 2020-03-11 09:30:00              | 2020-03-11 09:45:00             |
| 242       | Venise             | 2020-03-12 07:45:00              | 2020-03-12 08:00:00             |
| 242       | Belgrade           | 2020-03-13 19:40:00              | 2020-03-13 19:55:00             |
| 242       | Sofia              | 2020-03-14 15:45:00              | 2020-03-14 16:00:00             |
| 422       | Paris              | 2020-03-10 11:45:00              | 2020-03-10 12:00:00             |
| 422       | Lausanne           | 2020-03-11 07:45:00              | 2020-03-11 08:00:00             |
| 422       | Milan              | 2020-03-11 20:45:00              | 2020-03-11 21:00:00             |



## Exercice 13

MCD :



MPD (MLD indisponible sur PowerAMC) :



# TD 5

## Modification de l'ancienne requête :

```
drop table if exists COMPTE_RENDU;
drop table if exists ELEVE;
drop table if exists PROF;
drop table if exists UTILISATEUR;

create table COMPTE_RENDU
(
    NUM_USER          int          not null,
    NUM_CR            int          not null,
    DESCRIPTION_CR    char(100),
    DATE_CR           Date,
    primary key (NUM_USER, NUM_CR)
)
ENGINE = InnoDB;

create index FAIRE_FK on COMPTE_RENDU
(
    NUM_USER
);
create table ELEVE
(
    NUM_USER          int          not null,
    `OPTION`         char(10),
    primary key (NUM_USER)
)
ENGINE = InnoDB;
create table PROF
(
    NUM_USER          int          not null,
    primary key (NUM_USER)
)
ENGINE = InnoDB;
create table UTILISATEUR
(
    NUM_USER          int          not null,
    NOM_USER          char(50),
    PRENOM_USER       char(50),
    LOGIN_USER        char(50),
    MDP_USER          char(100),
    EMAIL_USER        char(100),
    primary key (NUM_USER)
)
ENGINE = InnoDB;
alter table COMPTE_RENDU add constraint FK_FAIRE foreign key (NUM_USER)
references ELEVE (NUM_USER) on delete restrict on update restrict;
alter table ELEVE add constraint FK_HERITAGE_1 foreign key (NUM_USER)
references UTILISATEUR (NUM_USER) on delete restrict on update restrict;
alter table PROF add constraint FK_HERITAGE_2 foreign key (NUM_USER)
references UTILISATEUR (NUM_USER) on delete restrict on update restrict;
```

## Vérification de la BDD :

**DESCRIBE** COMPTE\_RENDU;

| Field          | Type      | Null | Key | Default | Extra |
|----------------|-----------|------|-----|---------|-------|
| NUM_USER       | int(11)   | NO   | PRI | NULL    |       |
| NUM_CR         | int(11)   | NO   | PRI | NULL    |       |
| DESCRIPTION_CR | char(100) | YES  |     | NULL    |       |
| DATE_CR        | date      | YES  |     | NULL    |       |

**SHOW CREATE TABLE** ELEVE;

| Table | Create Table                                           |
|-------|--------------------------------------------------------|
| ELEVE | CREATE TABLE `eleve` (<br>`NUM_USER` int(11) NOT NU... |

## Conclusion

Ce TP nous a introduit aux concepts avancés de la modélisation relationnelle, les associations ternaires, les associations réflexives, l'héritage, les relations multiples et les identifiants relatifs. Il nous a permis de traduire rigoureusement ces règles conceptuelles en modèles physiques (MLD) et en scripts SQL, garantissant ainsi l'intégrité de la base de données finale.